

Why this coordinate

Any autonomous scalar flow $d\bar{a}/d\bar{w} = g(\bar{a})$ is straightened by $u = \int d\bar{a}/g(\bar{a})$. Here $g = (1 - \bar{a})^2(1 + \delta a)$ gives $u = 1/(1 - \bar{a})$ and $du/d\bar{w} = 1 + \delta a$. The height writing's law is NOT autonomous ($\bar{a}'$ depends explicitly on $\bar{w}$), so it has no such coordinate; the state writing's defect and its remedy are two sides of the same property.

Physically $u = (1 + \delta a)(\bar{w} - \bar{w}_0)$ is the gap to the implied wall. The state changes from “how strong is the gain” to “how far am I”, and distance propagates by exactly how far one moved.

What it buys

1. The one-signed quadrature ratchet is removed IDENTICALLY, not reduced. $du/d\bar{w}$ is constant in $\bar{w}$, so the one-step integral is exact. Measured on the canonical command trajectory (same Σ , $\bar{a}$ -Euler versus exact- u): +1.641 % at the end of the descent, +2.122 % at the end of the oscillation, strictly one-signed throughout, and exactly first order in Δt (+1.077 / + 0.542 / + 0.272 % at $\Delta t/2, /4, /8$). Closed form: the $\bar{a}$ -Euler step equals $u[k+1] = u[k] + \Sigma + \Sigma^2/u + \mathcal{O}(\Sigma^3)$, and $\Sigma^2/u > 0$ regardless of direction — that is the ratchet. +2.1 % is the right size against the 2.8 point gap between arm (a) at 4.68 % and the true-slope arm at 1.87 %.
2. The predict increment reads $\hat{u}$ nowhere: $du = (1 + \delta a)d\bar{w}$. That is the property the “ $\bar{a}'$ from the TRUE height” arm had, obtained here without knowing the true height.
3. u is affine in the unknown wall position, so a Gaussian posterior is closer to the true posterior shape in u than in $\bar{a}$. An EKF in u is a DIFFERENT, non-equivalent estimator — this is the mechanism by which it can win.
4. $(u, \delta a)$ separate cleanly: a wall-position error is a constant offset in u , a δa error is a ramp proportional to accumulated displacement ($F_e(4, 5) = \hat{\Sigma}$).

What it does not buy

No information. $u = (1 + \delta a)(\bar{w} - \bar{w}_0)$ with $\bar{w}$ commanded and free, so $(u, \delta a) \leftrightarrow (\bar{w}_0, \delta a)$ is a known bijection at fixed $\bar{w}$, and Fisher information is invariant under smooth reparametrisation. What changes is conditioning. (That invariance assumes the prior is pushed forward consistently — see the seed section.)

The self-amplification is NOT abolished. The term $-2(1 - \hat{\bar{a}})\Sigma$ is exactly the coordinate's own time-varying Jacobian: a fixed e_u really does correspond to an $e_{\bar{a}}$ that grows 123× between $\bar{h} = 22.2$ and 2.0. What changes is that it moves from the propagation Jacobian to the output map $\partial\bar{a}/\partial u = u^{-2}$, where it does not compound.

The structural limits of the $\bar{a}$ writing survive unchanged: in a hold $\hat{\Sigma} = 0$ so $F_e(4, 5) = 0$ and δa is unobservable through the state path; over a closed oscillation $\sum \hat{\Sigma} = 0$ so the state path carries zero net information about it. Only y_2 sees δa continuously.

Also unchanged: the y_1 gain-correction path through $K_1 = PH_1^\top/S_1$ (measured to supply essentially all of the across-seed spread, and to be correctly priced at 0.04–16 % of the realised optimum); the MA(2) augmentation the controller runs but no derivation describes (≈ 21 % of the spread); $\bar{h}_{\text{safe}} = 1.5$ as an underived house value.

Seed, prior, and a well-posedness constraint

δa enters the LAW here, so its prior enters u at $k = 0$ and $P_{u\delta a}[0] \neq 0$. With $\hat{u}[0] = 22.22$ and $\sqrt{P_{\delta a}} = 0.5$ the three contributions to $P_{uu}[0]$ are 123.5 (δa), 0.012 (wall position) and $\hat{u}[0]^4 \text{floor}_a^2$; the disturbance term dominates the wall term by 10^4 .

floor_a MUST be re-derived for THIS family. The single-curve family of `formC_state_dist` declares 13.05 % RMS / 15.34 % sup; this family, which carries δa , is the one declared in `formC_state_da` at 0.66 % / 4.65 %. The choice matters far more here than in $\bar{a}$ coordinates because the floor is amplified by $\hat{u}[0]^4$:

floor_a	σ_u	$P(u < 1)$	status
0.0306 (envelope sup)	15.11	8.0 %	ill-posed
0.0056 (local at seed)	2.77	≈ 0	well-posed

$u > 1 \Leftrightarrow \bar{a} < 1$, and $u \rightarrow 1$ is the singularity of the output map $1 - 1/u$. Eight percent of prior mass at $\bar{a} < 0$ sitting on that singularity is not a width question. In $\bar{a}$ coordinates the same two choices differ only by a factor 5.5 in width; a linear push-forward cannot preserve both tails. The implementation therefore uses the local floor and clamps $u \geq u_{\min}$ (with $\bar{h}_{\text{safe}} = 1.5$ the natural value is $u_{\min} = 1.5$), and clamps $\hat{v}$ likewise.

Reporting $\hat{\bar{a}} = 1 - 1/\hat{u}$ carries a Jensen bias $\approx -P_{uu}/\hat{u}^3$: negligible once converged (0.01 % at $u = 2$, 0.03 % at $u = 4.2$) but 2.18 % at the seed under the sup floor versus 0.07 % under the local floor — the same choice again.

Verification, 2026-08-13

Independent check before implementation. Central claim, S6, S7 and S8 confirmed term by term, including a time-varying similarity transform $F^u[k] = T[k+1]F^{\bar{a}}[k]T[k]^{-1}$ with $T = \hat{u}^2$ that reproduces the printed F_e exactly. Three errors were found and are fixed in the main file:

1. $P_{uu}[0]$ omitted the δa term and $P_{u\delta a}[0]$ was set to zero. Inherited from the wrong sibling (see 3), where the zero is correct because δa does not enter the law there. Magnitude-dominant.
2. The delay back-off was labelled exact; it drops the $(\delta\bar{w}_3 - \delta\bar{w}_1)$ channel, which `formB_ws_ref` already declares as a truncation. Measured contribution 0.5–3 % of $\sqrt{R_2}$ (peak 11 %), i.e. ≤ 0.1 % of variance, but it is correlated with the states, so it is bias and not variance. Now declared. Note R_2 's $d\hat{v}^{-4}Q_{uu}$ exists precisely because H omits that channel, so “exact” and that term cannot both stand.
3. The sibling cross-reference pointed at `formC_state_dist.tex`, whose δa is a different object (additive per step, absent from the law). The $\bar{a}$ -coordinate version of THIS law is `formC_state_da.tex`.

Also reported: `formC_state_da.tex` is internally inconsistent on Q_{33} (prefactor $4k_B T/(a_o R)$ in one place, $4k_B T a_o/R$ in its ξ); the main file here is the correct one.

Pre-registration

On the canonical scenario over the 8 house seeds, overall relative gain-error RMS: the same estimator in $\bar{a}$ coordinates gives $4.68 \pm 2.56\%$; fed $\bar{a}'$ from the true height it gives $1.87 \pm 0.93\%$; production tier-2 gives $2.07 \pm 1.38\%$.

lands near 4.7 the gap is information; the coordinate cannot help
lands near 1.9 the gap is conditioning
lands below 1.9 also expected: the true-slope arm still carries the
 Euler ratchet, which this writing removes as well

Arm (a) against 4.68% is the clean controlled comparison: same law, same estimator, same measurement chain, differing only in the quadrature and in whether the integrand reads the state.

Result, 2026-08-13

Eight house seeds, paired, canonical scenario. Overall relative gain-error RMS:

	desc peak	osc RMS	overall RMS
$\bar{a}$ coordinate, arm (a)	15.25 ± 6.30	4.24 ± 1.74	4.59 ± 2.56
u coordinate, arm (a)	16.02 ± 7.70	4.42 ± 2.07	5.46 ± 2.90
$\bar{a}$ coordinate, arm (b)	17.52 ± 6.41	7.39 ± 7.32	7.33 ± 4.61
u coordinate, arm (b)	17.09 ± 4.80	4.22 ± 2.09	4.76 ± 2.71
$\bar{a}'$ from the true height	3.23 ± 0.98	1.58 ± 1.01	1.87 ± 0.93
production tier-2	4.22 ± 1.80	1.90 ± 1.00	2.07 ± 1.38

Arm (a) paired difference $u - \bar{a}$: $+0.874 \pm 1.605$, $t = +1.54$, u better on 3/8. THE FIRST PRE-REGISTERED BRANCH: it lands near 4.7, not near 1.9. The coordinate does not close the gap, and the point estimate is on the wrong side of zero. The $2.4\times$ gap to production tier-2 is INFORMATION, not conditioning.

That is decisive because the ratchet removal is not in doubt — it was measured at $+2.1\%$, one-signed, and this writing removes it identically. Removing it buys no accuracy. So the ratchet was already being corrected by the measurements, and what survives is what the true-slope arm removes and this coordinate cannot: the integrand reads the state, and no smooth reparametrisation changes what the data say about the state. The invariance argument of “What it does not buy” is thereby confirmed by experiment rather than only asserted.

Read off the figures: the error trace has the SAME SHAPE in both coordinates — same features at the same times — shifted bodily. Seed 7 shifts the wrong way (final hold $-3 \rightarrow -5\%$), seed 101 the right way (final hold returns toward zero). A shift whose sign is set by the seed, not by the coordinate, is what $t = +1.54$ over 3/8 means.

Two findings that are not about the headline:

1. In arm (b) the coordinate HELPS: $7.33 \rightarrow 4.76$, u better on 7/8 ($t = -1.71$). And $\delta\hat{a}$ actually moves here — -0.145 ± 0.058 , one-signed 8/8 — where in `formC_state_dist` it never left noise

(-5.4×10^{-8}) . The difference is structural: δa enters the law in this family, so $F_e(4, 5) = \hat{\Sigma}$ makes it observable through displacement. The disturbance state is only estimable in the writing where it means something. It still does not reach arm (a), so this rehabilitates the mechanism without rehabilitating the arm.

2. The well-posedness table was right, and it was worth having. The envelope floor gives 15.32 % overall RMS against the local floor's 5.46 %, and clamps u on up to 23.5 % of steps against 4.0 %. Predicted before the run from $P(u < 1) = 8.0$ % alone.

The clamp is not the explanation for the headline: per-seed clamp count against the paired difference correlates at -0.008 .

Standing after this result

Three independent routes have now failed to close the same gap — an additive disturbance state, a multiplicative one, a uniform-in-time covariance inflation, and a coordinate change that provably removes the discretisation error. The one thing that does close it is knowing where the wall is. Production tier-2 does not know either, and matches the true-slope arm, so the remaining question is not how to integrate the gain but whether anything in the state writing can substitute for evaluating $\bar{a}'$ at a height. On the present evidence the answer is no, and the state writing's case has to rest on something other than accuracy on this scenario — extrapolation beyond the traversed band being the obvious candidate, and still unmeasured.